

Microeconomics

Lecture 18: Oligopoly and Game Theory

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What we know

So far we have studied two extreme cases of market structure:

- **Perfect competition:** many firms, $P = MC$, $\pi = 0$ in the long run, maximum efficiency
- **Monopoly:** a single firm, $P > MC$, $\pi > 0$ in the long run, significant deadweight loss

Reality, however, almost always lies in between. Most markets are dominated by a *small number* of firms: smartphones, automotive, airlines, large-scale retail, oil, salty snacks.

What we will see

The fundamental feature of oligopoly is **strategic interdependence**: each firm's profit also depends on its competitors' choices. To analyze it we need the tools of *game theory*. We will then study the three classic models (Bertrand, Cournot, Stackelberg) and the problem of collusion.

Learning objectives

1. **Define** a game in normal form and the concepts of dominant strategy and Nash equilibrium
2. **Identify** the Nash equilibrium in a payoff matrix
3. **Recognize** the structure of the prisoner's dilemma and apply it to oligopoly
4. **Compute** the Bertrand paradox and discuss the ways out (differentiation, capacity, collusion)
5. **Derive** the sustainability condition for collusion using repeated games and geometric series
6. **Compute** the Cournot equilibrium (including with asymmetric costs)
7. **Compute** the Stackelberg equilibrium and compare it with Cournot

Reference market for the whole lecture

$$P = 30 - Q = 30 - q_1 - q_2, \quad MC_1 = MC_2 = 6, \quad FC = 0.$$

Known benchmark equilibria: monopoly $Q^M = 12$, $P^M = 18$, $\pi^M = 144$. Perfect competition: $Q_{pc} = 24$, $P_{pc} = 6$, $\pi = 0$.

Oligopoly in real-world markets

Oligopoly

A market structure with a **small number** of firms (typically 2–10), each with *significant market power* and aware that its choices affect its competitors.

Real-world examples (concentration of the top 3–5 firms):

- **Smartphones (global):** Apple, Samsung, Xiaomi > 50% of the market
- **Automotive (Europe):** Volkswagen, Stellantis, Renault, BMW, Mercedes $\approx$ 75%
- **Airlines (Italy):** typically 2–4 carriers per route (Ryanair, ITA, Lufthansa, easyJet)
- **Large-scale retail (Italy):** Conad, Coop, Selex, Esselunga, Carrefour, Lidl $\approx$ 70%
- **Private-label salty snacks (Italy):** Amica Chips, Pata, Preziosi Food (more on this later)
- **Oil (world):** OPEC+ controls about 40% of production

Concentration does not automatically imply abuse, but it makes the market work *differently* from both perfect competition and pure monopoly.

Strategic interdependence

What is new compared with PC and monopoly

In oligopoly, each firm's optimal decision **depends** on the decisions of the others. It is no longer enough to maximize profit given market demand: firms must also *anticipate* their competitors' moves.

Consequence. None of the techniques seen so far is sufficient:

- Under PC the firm is a price-taker: it takes P as given.
- Under monopoly the firm faces the *entire* market demand.
- Under oligopoly the firm faces a *residual demand* that depends on how much/how the others price.

The tool

We need concepts that describe strategic interaction: *strategy*, *best response*, *equilibrium*. This is what **game theory** provides.

What is a game

Game in normal form

A **game in normal form** is defined by three elements:

1. **Players:** who takes part ($i = 1, \dots, n$)
2. **Strategies:** the set of actions available to each player (S_i)
3. **Payoffs:** the function assigning an outcome to each combination of strategies ($u_i(s_1, \dots, s_n)$)

Standard assumptions:

- Players are **rational**: they maximize their own payoff
- Players are **strategic**: they anticipate the others' choices
- The rules of the game are **common knowledge**: everyone knows that everyone is rational, and everyone knows that everyone knows it

Representation. Games with two players and discrete strategies are represented by a **payoff matrix**: rows = player A, columns = player B, each cell contains the pair (u_A, u_B) .

Dominant strategy

(Strictly) dominant strategy

A strategy s_i^* is **strictly dominant** for player i if it is his or her optimal choice whatever the opponent does:

$$u_i(s_i^*, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall s_i \neq s_i^*, \forall s_{-i}$$

Example: the advertising game. Two firms A and B decide whether to advertise (profits in millions of €):

	B: Advertise	B: No advertising
A: Advertise	(3, 3)	(5, 1)
A: No advertising	(1, 5)	(4, 4)

Analysis for A. If B advertises, A prefers to advertise ($3 > 1$). If B does not, A still prefers to advertise ($5 > 4$). Hence **“Advertise” is dominant for A** (by symmetry, also for B).

Outcome. Both advertise $\Rightarrow$ payoffs (3, 3). Yet (4, 4) would be Pareto-superior!

Nash equilibrium

Nash equilibrium

A **Nash equilibrium** (NE) is a strategy profile $(s_1^*, \dots, s_n^*)$ such that no player has an incentive to deviate **unilaterally**:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i, \forall i$$

Intuition. In an NE everyone does the best they can *given* what the other does. Nobody regrets their choice once they observe the other's.

Procedure for finding it in the matrix (best-response highlighting):

1. For each column, underline the row player's maximum payoff
2. For each row, underline the column player's maximum payoff
3. Cells with *both* payoffs underlined are Nash equilibria

Nash $\supset$ Dominant

If both players have a dominant strategy, the NE coincides with the combination of dominant strategies. But Nash is more general: it exists even when there are no dominant strategies.

Finding the Nash equilibrium: the method applied

Same advertising game, applying the method. I underline A's maximum payoff in each column, and B's maximum payoff in each row:

	B: Advertise	B: No advertising
A: Advertise	(<u>3</u> , <u>3</u>)	(<u>5</u> , 1)
A: No advertising	(1, <u>5</u>)	(4, 4)

Counting the underlines.

- Column 1 (B: Advertise): A's max is 3 at the top $\Rightarrow$ underline. Column 2: max is 5 at the top $\Rightarrow$ underline.
- Row 1 (A: Advertise): B's max is 3 on the left $\Rightarrow$ underline. Row 2: max is 5 on the left $\Rightarrow$ underline.

Cell with *both* underlined: only (Advertise, Advertise). **Unique NE** = (3,3).

The dilemma

(No, No) = (4, 4) is Pareto-superior, but **it is not an equilibrium**: each firm has an incentive to deviate unilaterally to "Advertise".

The prisoner's dilemma

The most famous game in economics: it shows why rational individuals can reach a *collectively suboptimal* outcome.

The original story. Two criminals are arrested separately and held in cells with no communication. Each can *confess* (betray the accomplice) or *stay silent* (cooperate with the accomplice). The prosecutor makes the same offer to both (years in prison, negative payoffs):

	Accomplice: Silent	Accomplice: Confesses
Suspect: Silent	$(-1, -1)$	$(-10, 0)$
Suspect: Confesses	$(0, -10)$	$(-5, -5)$

Analysis. “Confess” is a dominant strategy for both (check it!): if the other stays silent, $0 > -1$; if the other confesses, $-5 > -10$. NE: (Confess, Confess) $\Rightarrow (-5, -5)$.

The paradox

The equilibrium outcome is *Pareto-dominated* by (Silent, Silent) = $(-1, -1)$. Individual rationality leads to a collectively worse outcome.

The prisoner's dilemma applied to oligopoly

Translation into oligopoly language. Two firms choose the price: *high* (= behave like a cartel, cooperating with the rival) or *low* (= price war, betraying the rival).

(profits in thousands of €)	Firm 2: high P	Firm 2: low P
Firm 1: high P	(8, 8)	(2, 12)
Firm 1: low P	(12, 2)	(4, 4)

Conceptual mapping:

Prisoner	$\longleftrightarrow$	Oligopoly
Silent (cooperates with accomplice)	$\longleftrightarrow$	High price (abides by cartel)
Confesses (betrays)	$\longleftrightarrow$	Low price (price war)
(Silent, Silent) Pareto-optimal	$\longleftrightarrow$	Collusion: high profits
NE (Conf., Conf.)	$\longleftrightarrow$	Price war: low profits

Same structure, same result. “Low price” is dominant for both $\Rightarrow$ NE: (4, 4). The collusive optimum (8, 8) is *not* an equilibrium: each firm deviates unilaterally, because $12 > 8$.

The Bertrand model

Joseph Bertrand (1883) proposed a model in which firms compete on **prices** rather than quantities.

Assumptions

- Two firms, **homogeneous good**, same $MC = 6$, **unlimited capacity**
- They choose prices P_1 and P_2 **simultaneously**
- Consumers buy from the firm with the *lowest* price (all demand goes there)
- In case of a tie $P_1 = P_2$: demand is split in half

Reference market. $P = 30 - Q$, $MC = 6$. If a single firm served the whole market at a price P , the quantity demanded would be $Q(P) = 30 - P$.

Key question

What price would you set, if you were one of the two firms, knowing that your rival will reason the same way?

Price competition

Imagine you are one of the two competing firms. Both firms have $MC = 6$ and unlimited capacity. Market demand is $Q = 30 - P$.

Think of an **integer price** P at which you would sell. The winner takes most of the market, or the market is split in half.

What price do you choose?

What typically happens:

- Someone proposes $P = 18$ (shared monopoly): profit $\pi = (18 - 6) \cdot 6 = 72$ each
- Someone else thinks: “if the other proposes 18, at 17 I take everything $\Rightarrow 11 \cdot 13 = 143$ ”
- Someone else: “then I go down to 16...” then 15, 14, 13...
- Where does it stop?

Bertrand: the logic of undercutting

Let us formalize the classroom reasoning.

Case 1. $P_2 > MC = 6$. Firm 1 has an obvious incentive: if it sets $P_1 = P_2 - \varepsilon$ (with ε very small) it captures *all* demand at price P_1 with a positive margin:

$$\pi_1 = (P_2 - \varepsilon - 6) \cdot (30 - (P_2 - \varepsilon)) > \frac{(P_2 - 6)(30 - P_2)}{2}$$

so it pays to “undercut” the rival.

Case 2. Firm 2 responds with the same logic: $P'_2 = P_1 - \varepsilon$. And so on.

Where does the spiral stop? At no price $P > MC$ is there an equilibrium: each firm would like to go just below the other. The only “fixed point” is $P_1 = P_2 = MC = 6$:

- Nobody can go lower (they would sell below cost: negative profits)
- Nobody can raise the price unilaterally (they would lose all customers)

The Bertrand paradox

Bertrand equilibrium

$$P_1^* = P_2^* = MC = 6, \quad \pi_1 = \pi_2 = 0$$

Just two firms replicate the perfectly competitive outcome.

Why “paradoxical”?

- Going from 1 firm (monopoly) to 2 firms makes profits *collapse* from 144 to 0
- Going from 2 to 3 firms (or 100) changes nothing: profits are already zero
- Empirically this behavior is *not observed*: real duopolies rarely set $P = MC$

The key to the paradox

The model is extremely sensitive to three assumptions:

1. Perfectly homogeneous good
2. Unlimited production capacity of each firm
3. One-shot (non-repeated) interaction

Let us relax them one at a time.

Way out 1: product differentiation

If the good is not perfectly homogeneous, consumers do not instantly shift demand to the lowest price. Each firm faces its *own* residual demand even with $P_i > P_j$.

Example: two nearby cafés

Café A next to our classroom, Café B 200 meters further on, near the bus stop. Both sell coffee at $MC \approx 0.40$ €. If Café A prices coffee at 1.10 € and Café B at 1.20 €:

- Students leaving here go to the nearest one (Café A): few walk 200 meters to save 10 cents
- Those taking the bus go to the nearest one (Café B) for the same reason
- Both cafés keep customers and prices above marginal cost

Hotelling model (intuition). The distance (d) between firms acts as a transport cost $t \cdot d$ borne by the consumer. Equilibrium with prices above MC :

$$P_i^* = MC + t \cdot d$$

Differentiation (spatial or product) creates a *margin*: it is the opposite of the paradox.

Way out 2: capacity constraints

If no firm can serve *all* demand at the price $P = MC$, Bertrand's argument breaks down: even by setting a very low price, the undercutter does *not* capture the whole market.

Intuition

Suppose firm 1 has maximum capacity $\bar{q}_1 = 15$ (while total demand at $P = MC$ would be $Q = 24$). If it sets P_1 just below P_2 , it sells $\bar{q}_1 = 15$. The remaining 9 consumers still go to firm 2, which can therefore set a price $P_2 > MC$ by serving the *residual* demand.

Edgeworth model (1897). With constrained capacity, a pure-strategy equilibrium often *does not exist*: prices oscillate between a competitive minimum and a monopoly maximum (Edgeworth cycle).

Real-world cases. Capital-intensive sectors with rigid capacity (steel, cement, electricity, aviation). This explains why in these markets prices do not collapse to MC even with few operators.

Way out 3: the game is repeated

Third critical assumption: the game is *one-shot*.

In real markets, however, firms meet **many times**, every day, for years. This radically changes the analysis.

Intuition

If I deviate from collusion today, I earn more *today*, but if the other firm can react *tomorrow*, I lose profits in all future periods. If future periods “matter enough”, the incentive to deviate disappears.

This is the most important way out, both theoretically and for antitrust policy. The next slides develop it formally:

1. Collusive calculation in the reference market
2. Mathematical tool: geometric series
3. Punishment strategy: grim trigger
4. Sustainability condition on the discount factor δ

Collusion: the incentive to coordinate

Collusion

Collusion occurs when firms coordinate their choices to maximize *joint* profit, behaving like a shared monopoly.

In the reference market. $P = 30 - Q$, $MC = 6$. The joint monopoly optimum:

$$Q^M = 12, \quad P^M = 18, \quad \pi^M = (18 - 6) \cdot 12 = 144$$

Equal split between the two firms:

$$q_1^{\text{coll}} = q_2^{\text{coll}} = 6, \quad \pi_1^{\text{coll}} = \pi_2^{\text{coll}} = \frac{144}{2} = 72$$

Comparison

- One-shot Bertrand: $\pi_i = 0$
- Collusion: $\pi_i = 72$

If the firms managed to sustain the agreement, they would go from zero profits (competitive duopoly) to 72 each, appropriating the maximum possible profit in the market.

The incentive to deviate

But collusion is unstable. Firms compete on price (Bertrand). Suppose firm 2 abides by the agreement and sets the collusive price $P_2 = 18$. Firm 1 asks itself: “should I abide by it too?”

The logic of undercutting. It is exactly the reasoning of the Bertrand paradox: if firm 1 sets $P_1 = 18 - \varepsilon$ (with ε very small), it captures the *entire* market.

$$P_1 = 18 - \varepsilon \Rightarrow Q_1 = 30 - 18 = 12, \quad P_2 = 18 \Rightarrow Q_2 = 0$$

Profits if 1 deviates (and 2 abides), in the limit $\varepsilon \rightarrow 0$:

$$\pi_1^{\text{dev}} = (18 - 6) \cdot 12 = 144, \quad \pi_2 = 0$$

By deviating, firm 1 appropriates the *entire monopoly profit*: it sells at (almost) the collusive price, but to the whole market.

The key point

$\pi_1^{\text{dev}} = 144 \gg 72 = \pi_1^{\text{coll}}$. **Deviating pays off hugely.** In Bertrand the deviator takes everything: by symmetry it also pays for firm 2. In a *one-shot* game collusion is *not* a Nash equilibrium.

Repeated game: a new time horizon

Change of perspective. Firms do not meet only once: they interact in every period $t = 0, 1, 2, \dots$

Discount factor

The **discount factor** $\delta \in (0, 1)$ measures how much future profits are worth today. A profit π earned in period t is worth $\delta^t \pi$ today.

High $\delta \Leftrightarrow$ *patient* firms, the future weighs a lot. $\delta \approx 1/(1 + r)$ with r the interest rate.

Two types of repeated game:

- **Finite horizon** (T known periods). By *backward induction*: in the last period nobody punishes anymore, so everyone deviates; knowing this, everyone already deviates in period $T - 1$; and so on backward. Result: Cournot/Bertrand is played in every period. Collusion does *not* hold.
- **Infinite (or indefinite) horizon**. Without a “last period” the future punishment is always credible. Collusion *can* hold, if δ is high enough.

We will study the infinite-horizon case. Mathematical tool needed: *geometric series*.

Mathematical tool: geometric series

For discounted profits over an infinite horizon we will use this family of formulas.

Basic formula (for $|\delta| < 1$):

$$\sum_{t=0}^{\infty} \delta^t = 1 + \delta + \delta^2 + \delta^3 + \dots = \frac{1}{1 - \delta}$$

Notable cases (derived from the basic formula):

Series	Sum	When it is used
$1 + \delta + \delta^2 + \dots$	$\frac{1}{1 - \delta}$	constant flow from <i>today</i>
$\delta + \delta^2 + \delta^3 + \dots$	$\frac{\delta}{1 - \delta}$	flow from <i>tomorrow</i> (grim trigger)
$\delta^k + \delta^{k+1} + \dots$	$\frac{\delta^k}{1 - \delta}$	flow starting in k periods
$1 + \delta^k + \delta^{2k} + \dots$	$\frac{1}{1 - \delta^k}$	flow only every k periods
$1 + \delta + \dots + \delta^{T-1}$	$\frac{1 - \delta^T}{1 - \delta}$	finite horizon of T periods

The grim trigger strategy

To sustain collusion, a credible threat of punishment is needed.

“Grim trigger” strategy

“I cooperate (set the collusive price $P^{\text{coll}} = 18$) as long as the other cooperates. As soon as I observe a deviation, I go back to competing à la **Bertrand forever** ($P = MC = 6$, zero profits in every subsequent period).”

Comparison of the two scenarios for firm 1, from period $t = 0$.

Scenario A: I abide by the agreement. I obtain $\pi^{\text{coll}} = 72$ in every period:

$$V_A = 72 + 72\delta + 72\delta^2 + \dots = \frac{72}{1 - \delta}$$

Scenario B: I deviate today. Today I undercut and earn $\pi^{\text{dev}} = 144$ (the whole market). From tomorrow the rival punishes $\Rightarrow$ back to Bertrand $\Rightarrow$ I earn $\pi = 0$ forever:

$$V_B = 144 + 0 \cdot \delta + 0 \cdot \delta^2 + \dots = 144$$

Sustainability condition (derivation)

Collusion is sustainable if abiding by the agreement pays at least as much as deviating: $V_A \geq V_B$.

$$\frac{72}{1 - \delta} \geq 144$$

Multiply both sides by $(1 - \delta) > 0$:

$$72 \geq 144(1 - \delta) = 144 - 144\delta$$

$$144\delta \geq 72 \implies \boxed{\delta \geq \delta^* = \frac{72}{144} = \frac{1}{2}}$$

Interpretation

Collusion is sustainable if firms are **sufficiently patient**: the discount factor must be at least $1/2$. With normal interest rates ($\delta = 1/(1 + r)$ close to 0.95) the condition is easily satisfied: the real problem for cartels is *the observability of deviations*, not patience.

General formula. $\delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}}$, where π^{punish} is the punishment profit.

What if δ were different?

Firms compete à la Bertrand in the reference market ($P = 30 - Q$, $MC = 6$): $\pi^{\text{coll}} = 72$, $\pi^{\text{dev}} = 144$ (the deviator captures the whole market), $\pi^{\text{punish}} = 0$. The punishment strategy is grim trigger.

(a) Is collusion sustainable with $\delta = 0.4$?

(b) And with $\delta = 0.7$?

(c) For which value of δ are firms *indifferent* between deviating and cooperating?

(a) $\delta = 0.4$. Compare with $\delta^* = 1/2$. Since $0.4 < 0.5$, collusion is *not* sustainable.

$$V_A = \frac{72}{1-0.4} = \frac{72}{0.6} = 120, \quad V_B = 144$$

$V_B > V_A$: deviating pays.

(b) $\delta = 0.7$. Since $0.7 > 0.5$, collusion is sustainable.

$$V_A = \frac{72}{1-0.7} = \frac{72}{0.3} = 240, \quad V_B = 144$$

$V_A > V_B$: abiding by the agreement pays.

(c) Indifference occurs exactly at $\delta = \delta^* = 1/2$ ($V_A = 72/0.5 = 144 = V_B$).

Generalization: δ^* as a function of the number of firms

How does δ^* change if the cartel has n symmetric firms? Under Bertrand competition the calculation is immediate: collusion splits the monopoly profit ($\pi_i^{\text{coll}} = \pi^M/n$), deviation captures the whole market ($\pi^{\text{dev}} = \pi^M$), punishment drives profits to zero ($\pi^{\text{punish}} = 0$).

$$\delta^*(n) = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}} = \frac{\pi^M - \pi^M/n}{\pi^M - 0} = 1 - \frac{1}{n} = \frac{n-1}{n}$$

n	2	3	4	5	6	7	...
$\delta^* = (n-1)/n$	1/2	2/3	3/4	4/5	5/6	6/7	$\rightarrow 1$
value	0.500	0.667	0.750	0.800	0.833	0.857	$\rightarrow 1$

Reading

$\delta^*(n) = (n-1)/n$ is **increasing in n** and *does not depend* on demand or cost parameters. The more firms there are, the more patient they must be to sustain the cartel; as $n \rightarrow \infty$, $\delta^* \rightarrow 1$: with many firms collusion becomes practically unsustainable.

Real-world case: the potato chip cartel (AGCM, April 2026)

A collusion exactly like the one seen in theory, uncovered in Italy last month.

The facts established by the AGCM

The three main Italian producers of private-label salty snacks—**Amica Chips** (2023 turnover: €140 mn), **Pata** (171 mn), and **Preziosi Food** (86 mn)—coordinated their supply to large retail chains for over **eight years**.
Total fine: **€23.3 million**.

How the cartel worked.

- *Market sharing* by customer: Amica Chips → Esselunga, Lidl, Carrefour; Pata → Coop, Conad, Pam, Eurospin
- They exchanged *commercial terms and screenshots of offers* by email before responding to retailers' tenders
- Total Italian salty snack market: $\sim$ €922 mn, private-label segment $\sim$ 1/3

Mapping to the model.

- “Homogeneous good” → private-label chips (minimal differentiation)
- “Patient firms” → multi-year supply relationships $\Rightarrow$ high $\delta \Rightarrow$ sustainable collusion
- Modus operandi: *market sharing* (a variant of the standard collusive model)

How cartels break down: the meta prisoner's dilemma

If collusion is sustainable, how did the AGCM manage to uncover it?

The potato chip cartel was broken by a *whistleblower* (an employee or a participating firm that reported the cartel in exchange for a reduced fine).

Leniency program

The AGCM reduces the fine for whoever reports first: **full immunity** for reporting before the investigation is opened; decreasing reductions for those who cooperate later. In the chip case: Amica Chips and Pata obtained reductions for cooperating.

The meta-dilemma

The leniency program builds a *prisoner's dilemma* on top of the existing cartel:

- Stay silent (cooperate with the cartel): high profits but risk of detection
- Confess first to the AGCM: fine reduced to zero
- Equilibrium of the new game: **report before the other does**

The antitrust authority deliberately overturns the strategic structure in favor of competition.

What if firms competed on quantities instead of prices?

The Bertrand paradox depends on the assumption that the strategic variable is price. What happens if firms instead choose *quantities* (and the price emerges from market demand)?

When does this make sense? In markets where *production capacity* is the binding decision. Once capacity is installed, the price forms on the market to balance demand and supply.

Real-world examples of “Cournot-style” competition:

- **Steel and cement:** plant capacity decided years in advance, price adjusts
- **Electricity:** producers decide how much power to feed in, price forms on the power exchange
- **Raw materials** (copper, aluminum): extraction capacity fixed, price on the global market
- **OPEC+:** each country decides its own production quota, the Brent price adjusts

Augustin Cournot (1838) proposed the first formal model of oligopoly. We derive it in the next slides.

The Cournot model

Assumptions

- Two firms (duopoly): firm 1 and firm 2
- They choose quantities q_1 and q_2 **simultaneously**
- **Homogeneous** good: aggregate demand depends only on $Q = q_1 + q_2$
- Each firm maximizes its own profit taking the other's quantity as *given*

Reference market (the same throughout the lecture):

$$P = 30 - Q = 30 - q_1 - q_2, \quad MC_1 = MC_2 = 6, \quad FC = 0$$

Profit of firm 1:

$$\pi_1(q_1, q_2) = P \cdot q_1 - 6 q_1 = (30 - q_1 - q_2 - 6) q_1 = (24 - q_1 - q_2) q_1$$

By symmetry, $\pi_2 = (24 - q_1 - q_2) q_2$.

Reaction function

Firm 1 maximizes $\pi_1 = (24 - q_1 - q_2) q_1$ treating q_2 as given.

First-order condition:

$$\frac{\partial \pi_1}{\partial q_1} = 24 - 2q_1 - q_2 = 0 \implies q_1^*(q_2) = 12 - \frac{q_2}{2}$$

Reading the reaction (best-response) function. For each expected quantity q_2 , it gives firm 1's optimal quantity.

- If $q_2 = 0$ (no competition): $q_1^* = 12 = q^M$ —firm 1 becomes a monopolist
- If $q_2 = 24$ (maximum rival output, $P = MC$): $q_1^* = 0$ —it pays not to produce
- Slope $-1/2$: for each extra unit produced by the rival, firm 1 reduces its own output by half a unit

By symmetry, firm 2's reaction function:

$$q_2^*(q_1) = 12 - \frac{q_1}{2}$$

Cournot equilibrium

In equilibrium both firms choose optimally given their expectations about the other, and expectations are correct: each firm is on its own BR *at the same time*.

System:

$$\begin{cases} q_1 = 12 - \frac{q_2}{2} \\ q_2 = 12 - \frac{q_1}{2} \end{cases}$$

Solution. Substitute the second into the first:

$$q_1 = 12 - \frac{1}{2} \left(12 - \frac{q_1}{2} \right) = 12 - 6 + \frac{q_1}{4} = 6 + \frac{q_1}{4}$$

$$\frac{3q_1}{4} = 6 \implies \boxed{q_1^C = 8}$$

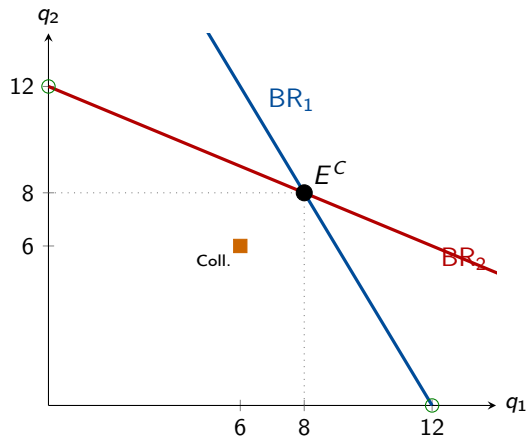
By symmetry: $q_2^C = 8$.

Total quantity, price, and profits:

$$Q^C = 16, \quad P^C = 30 - 16 = 14$$

$$\pi_1^C = \pi_2^C = (14 - 6) \cdot 8 = 8 \cdot 8 = 64$$

Graphical representation of the Cournot equilibrium



Reading.

The two BRs intersect at $E^C = (8, 8)$: the Cournot equilibrium.

Notable points:

- $(12, 0)$ and $(0, 12)$: individual monopoly
- $E^C = (8, 8)$: symmetric equilibrium
- $(6, 6)$: equal-split collusion

Collusion is *not* an equilibrium: starting from $(6, 6)$ each firm would like to move onto its own BR (e.g. firm 1 to $q_1 = 12 - 6/2 = 9$). Without enforcement mechanisms, collusion collapses toward E^C .

Collusion under Cournot: the incentive to deviate

Under Cournot too, firms could agree that each produces half the monopoly quantity: $q_i^{\text{coll}} = 6$ (so $Q = 12$, $P = 18$, $\pi_i = 72$). But is it an equilibrium?

Check: firm 1's best response to $q_2 = 6$. Unlike Bertrand, here the deviation is a choice of *quantity*. Firm 1 maximizes:

$$\pi_1 = (30 - q_1 - 6 - 6) q_1 = (18 - q_1) q_1 \Rightarrow \frac{\partial \pi_1}{\partial q_1} = 18 - 2q_1 = 0 \Rightarrow q_1^{\text{dev}} = 9$$

Market outcome if 1 deviates (and 2 abides):

$$Q = 9 + 6 = 15, \quad P = 15, \quad \pi_1^{\text{dev}} = (15 - 6) \cdot 9 = 81, \quad \pi_2 = 54$$

Comparison with the Bertrand deviation

$\pi_1^{\text{dev}} = 81 > 72 = \pi^{\text{coll}}$: collusion is *not* a Cournot equilibrium. But the gain from deviating is *much smaller* than under Bertrand (81 versus 144): competing in quantities, a firm cannot “steal” the rival's whole market, only expand its own output along its BR.

Sustainability of collusion under Cournot

With a repeated game and grim trigger, collusion also holds under Cournot if $\delta \geq \delta^*$. Here, however, the punishment is a return to the *Cournot equilibrium* ($\pi^C > 0$), not to Bertrand. For $n = 2$ in the reference market ($\pi^{\text{coll}} = 72$, $\pi^{\text{dev}} = 81$, $\pi^C = 64$):

$$\delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^C} = \frac{81 - 72}{81 - 64} = \frac{9}{17} \approx 0.529$$

Generalization to n firms (normalizing $(a - c)^2 = 1$: $\pi_i^{\text{coll}} = \frac{1}{4n}$, $\pi_i^C = \frac{1}{(n+1)^2}$, $\pi_i^{\text{dev}} = \left(\frac{n+1}{4n}\right)^2$):

n	2	3	4	...
δ^* (Cournot)	$9/17 \approx 0.529$	$4/7 \approx 0.571$	$25/41 \approx 0.610$	$\rightarrow 1$
δ^* (Bertrand)	$1/2 = 0.500$	$2/3 \approx 0.667$	$3/4 = 0.750$	$\rightarrow 1$

Reading

In both models δ^* is increasing in n and tends to 1. The comparison *between* the two models reflects **two opposing forces**: under Bertrand deviation pays more (the whole market is captured), and this *raises* δ^* ; but punishment is also harsher (zero profits versus the positive Cournot profits), and this *lowers* it. At $n = 2$ punishment narrowly prevails ($0.500 < 0.529$: collusion slightly easier under Bertrand); from $n \geq 3$ the temptation to deviate prevails.

Cournot vs. other models: summary table

Model	q_i	Q	P	π_i	Welfare
Monopoly (cartel)	6	12	18	72	low (high DWL)
Cournot ($n = 2$)	8	16	14	64	medium
Cournot ($n = 3$)	6	18	12	36	higher
Cournot ($n \rightarrow \infty$)	$\rightarrow 0$	$\rightarrow 24$	$\rightarrow 6$	$\rightarrow 0$	maximum (=PC)
Bertrand / Perfect competition	—	24	6	0	maximum

General result. With n identical firms and linear demand $P = a - bQ$, $MC = c$:

$$q_i^C = \frac{a - c}{(n + 1)b}, \quad Q^C = \frac{n(a - c)}{(n + 1)b}, \quad P^C = \frac{a + nc}{n + 1}$$

Convergence

For $n = 1$: monopoly. For $n \rightarrow \infty$: perfect competition. More firms $\Rightarrow$ more output, lower price, falling individual profits. **The number of firms is a continuous variable of competitive intensity.**

Cournot with asymmetric costs

Cournot duopoly with market demand $P = 120 - Q$.

Firm 1: $MC_1 = 20$. Firm 2: $MC_2 = 40$.

- (a) Write the profit functions π_1 and π_2 .
- (b) Derive the Cournot reaction functions.
- (c) Find the Cournot equilibrium (q_1^C, q_2^C) , Q^C , P^C .
- (d) Compute each firm's profits.
- (e) Comment on the result in terms of cost advantage.

Try it yourselves before looking at the solution. You have all the tools from the previous slides.

Solution

(a) Profit functions.

$$\pi_1 = (120 - q_1 - q_2 - 20) q_1 = (100 - q_1 - q_2) q_1$$

$$\pi_2 = (120 - q_1 - q_2 - 40) q_2 = (80 - q_1 - q_2) q_2$$

(b) Reaction functions. FOC for firm 1:

$$\frac{\partial \pi_1}{\partial q_1} = 100 - 2q_1 - q_2 = 0 \implies \boxed{q_1 = 50 - \frac{q_2}{2}}$$

FOC for firm 2:

$$\frac{\partial \pi_2}{\partial q_2} = 80 - q_1 - 2q_2 = 0 \implies \boxed{q_2 = 40 - \frac{q_1}{2}}$$

Remark. The BR intercepts (50 and 40) reflect the individual monopoly each firm would hold if the rival left the market.

Solution

(c) **Equilibrium.** Substitute BR_2 into BR_1 :

$$q_1 = 50 - \frac{1}{2} \left(40 - \frac{q_1}{2} \right) = 50 - 20 + \frac{q_1}{4} = 30 + \frac{q_1}{4}$$

$$\frac{3q_1}{4} = 30 \implies \boxed{q_1^C = 40}, \quad q_2^C = 40 - \frac{40}{2} = 20$$

$$Q^C = 60, \quad P^C = 120 - 60 = 60$$

(d) **Profits.**

$$\pi_1^C = (60 - 20) \cdot 40 = 1600, \quad \pi_2^C = (60 - 40) \cdot 20 = 400$$

(e) **Comment.** Firm 1 produces twice as much and *earns four times as much* as firm 2. The cost differential (20 vs 40) translates into a clear competitive advantage. Under Cournot **lower costs pay off twice**: producing more lowers the price, and at that price the efficient firm earns the larger margin.

What if one firm moved first?

So far firms have decided *simultaneously*. But in many markets one of the two moves *first*, and *observably* (irreversible commitment).

Real-world examples of timing asymmetry.

- **Tesla in electric cars (2008–2018):** invests in plants and gigafactories before VW, Stellantis, GM convert EV lines. Traditional carmakers react as followers
- **Ryanair vs traditional carriers:** fleet size and routes announced years in advance, Alitalia/Lufthansa/Air France adapt with the rest
- **OPEC+ vs non-OPEC+ producers:** OPEC+ announces production cuts/increases, the US and other non-OPEC+ producers react to the resulting price
- **Amazon vs traditional retailers in e-commerce:** logistics capacity installed first, the others play catch-up

Heinrich von Stackelberg (1934) modeled this situation. The key point: *moving first, if the commitment is credible, is a strategic advantage.*

The Stackelberg model

Assumptions

- **Firm 1 (leader):** chooses q_1 **first**, *observably* and *irreversibly*
- **Firm 2 (follower):** observes q_1 and then chooses q_2 optimally
- The leader knows that the follower will react with its BR and *takes* this reaction *into account*

Solution technique: backward induction. We start from the *second* stage (the follower) and work back to the first (the leader).

Stage 2 (follower). Given q_1 , firm 2 maximizes its profit. Same BR as in Cournot:

$$q_2^*(q_1) = 12 - \frac{q_1}{2}$$

Stage 1 (leader). Firm 1 knows that the follower will follow its BR. It substitutes $q_2 = 12 - q_1/2$ into its own profit and maximizes *with respect to* q_1 .

Stackelberg: computing the equilibrium

The leader substitutes the follower's BR:

$$\pi_1 = (24 - q_1 - q_2) q_1 = \left(24 - q_1 - \left(12 - \frac{q_1}{2}\right)\right) q_1 = \left(12 - \frac{q_1}{2}\right) q_1$$

Leader's FOC:

$$\frac{d\pi_1}{dq_1} = 12 - q_1 = 0 \implies \boxed{q_1^S = 12}$$

Follower's reaction:

$$q_2^S = 12 - \frac{12}{2} = 6$$

Market outcome:

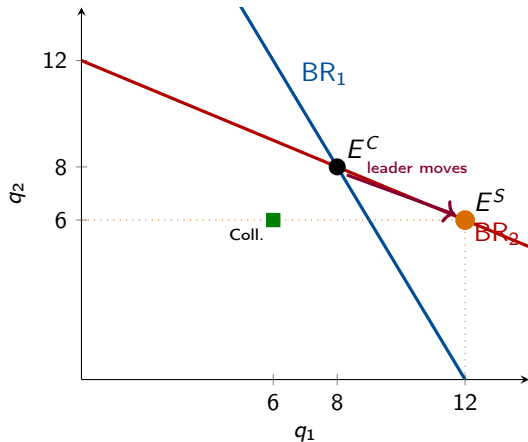
$$\begin{aligned} Q^S &= 12 + 6 = 18, & P^S &= 30 - 18 = 12 \\ \pi_1^S &= (12 - 6) \cdot 12 = 72, & \pi_2^S &= (12 - 6) \cdot 6 = 36 \end{aligned}$$

Key result

$\pi_1^S = 72 > 64 = \pi_1^C$: the leader earns more than under Cournot.

$\pi_2^S = 36 < 64 = \pi_2^C$: the follower earns less than under Cournot.

Stackelberg: graph



E^S lies on BR_2 : the follower makes its best move given $q_1 = 12$.

Why is $E^S \neq E^C$?

- Under Cournot *both* choose simultaneously, without being able to commit
- Under Stackelberg the leader credibly *commits* to $q_1 = 12$. The follower must adapt

The leader produces more than under Cournot ($12 > 8$): it “occupies” the market before the follower chooses. Geometrically, it chooses the point on the follower’s BR that is tangent to its own highest isoprofit curve.

Stackelberg: summary comparison

Reference market $P = 30 - Q$, $MC = 6$:

Model	q_1	q_2	Q	P	π_1	π_2
Monopoly (cartel)	6	6	12	18	72	72
Stackelberg	12	6	18	12	72	36
Cournot	8	8	16	14	64	64
Bertrand / PC	—	—	24	6	0	0

Key remarks.

- The leader produces more than under Cournot ($12 > 8$), the follower less ($6 < 8$)
- The leader earns as much as under collusion (72), the follower half that (36)
- Lower price than under Cournot ($12 < 14$): better for consumers
- The leader prefers Stackelberg ($72 > 64$); the follower prefers Cournot ($64 > 36$)

First-mover advantage

It exists *only if the commitment is observable and credible*. Without commitment, the only equilibrium is Cournot.

Asymmetric Stackelberg

Same market as the previous Cournot exercise: $P = 120 - Q$, $MC_1 = 20$, $MC_2 = 40$. Now firm 1 is the leader.

Stage 2 (follower). BR_2 from the FOC of $\pi_2 = (80 - q_1 - q_2)q_2$:

$$q_2^*(q_1) = 40 - \frac{q_1}{2}$$

Stage 1 (leader). Substitute q_2 into $\pi_1 = (100 - q_1 - q_2)q_1$:

$$\pi_1 = (100 - q_1 - 40 + q_1/2) q_1 = (60 - q_1/2) q_1$$

FOC: $60 - q_1 = 0 \Rightarrow \boxed{q_1^S = 60}$.

Follower's reaction: $q_2^S = 40 - 30 = 10$.

Outcome. $Q^S = 70$, $P^S = 50$, $\pi_1^S = (50 - 20) \cdot 60 = 1800$, $\pi_2^S = (50 - 40) \cdot 10 = 100$.

Cournot vs Stackelberg comparison (same market):

Model	q_1	q_2	π_1	π_2
Cournot	40	20	1,600	400
Stackelberg	60	10	1800	100

Exercise 1: prisoner's dilemma and sustainable collusion

Two duopoly firms decide whether to join an informal price agreement (C : cooperate) or compete aggressively for market share (D : deviate). Annual profits in millions of €:

	B: C	B: D
A: C	(4, 4)	(0, 6)
A: D	(6, 0)	(2, 2)

- (a) Are there dominant strategies? Find the NE of the one-shot game.
- (b) Does the NE coincide with the Pareto-optimal outcome?
- (c) The two firms meet *every year* with the same matrix. Using the grim trigger strategy, find the minimum value δ^* of the discount factor that makes cooperation (C, C) sustainable.

Exercise 1: solution

(a) Dominant strategies. For A: if B plays C , A prefers D ($6 > 4$); if B plays D , A still prefers D ($2 > 0$). D is a **dominant strategy** for A, and by symmetry also for B.

Unique one-shot NE: $(D, D) = (2, 2)$.

(b) Comparison with Pareto. $(C, C) = (4, 4)$ Pareto-dominates the equilibrium $(D, D) = (2, 2)$ but is *not* an NE: each firm would deviate unilaterally to D ($6 > 4$).

Prisoner's dilemma

$T = 6 > R = 4 > P = 2 > S = 0$.

It is a prisoner's dilemma: individual rationality $\Rightarrow$ collectively suboptimal outcome. Collusion (C, C) does not hold in a one-shot game.

The question becomes: *what changes if the firms meet every year?* (part c)

Exercise 1: solution

Repeated game with grim trigger. Firm A compares two strategies over the infinite horizon:

$$V_A^{\text{coll}} = 4 + 4\delta + 4\delta^2 + \dots = \frac{4}{1 - \delta}$$

$$V_A^{\text{dev}} = 6 + 2\delta + 2\delta^2 + \dots = 6 + \frac{2\delta}{1 - \delta}$$

Sustainability condition: $V_A^{\text{coll}} \geq V_A^{\text{dev}}$. Multiplying by $(1 - \delta) > 0$:

$$4 \geq 6(1 - \delta) + 2\delta \iff 4 \geq 6 - 4\delta \iff \boxed{\delta \geq \delta^* = \frac{1}{2}}$$

General formula (check).

$$\delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}} = \frac{6 - 4}{6 - 2} = \frac{1}{2} \quad \checkmark$$

Interpretation

It is enough for firms to give *at least half weight* to future profits (planning horizon of $\gtrsim 2$ years) to sustain collusion. With $\delta = 1/(1 + r)$ this is equivalent to $r \leq 100\%$, a condition easily met in real markets: the empirical problem of collusion is not patience, but the *observability* of deviations.

Exercise 2: symmetric Stackelberg

Duopoly in the market for industrial components. Market demand $P = 40 - Q$. Both firms have $MC = 4$, but firm 1 has already announced an irreversible investment plan that makes it the **leader**; firm 2 observes q_1 and then chooses q_2 .

- (a) Compute the equilibrium quantities q_1^S , q_2^S , Q^S , P^S by backward induction.
- (b) Compute the profits π_1^S and π_2^S .
- (c) Compare with the Cournot equilibrium in the same market. Which firm prefers Stackelberg, which Cournot? By how much?

Hint: start from the follower's FOC, then substitute into the leader's profit function.

Exercise 2: solution

Stage 2 (follower). $\pi_2 = (40 - q_1 - q_2 - 4) q_2 = (36 - q_1 - q_2) q_2$. FOC:

$$36 - q_1 - 2q_2 = 0 \implies q_2^*(q_1) = 18 - \frac{q_1}{2}$$

Stage 1 (leader). Substitute into the leader's profit function:

$$\pi_1 = \left(36 - q_1 - 18 + \frac{q_1}{2}\right) q_1 = \left(18 - \frac{q_1}{2}\right) q_1$$

FOC: $18 - q_1 = 0 \implies \boxed{q_1^S = 18}$.

Follower's reaction: $q_2^S = 18 - 9 = 9$. Hence $Q^S = 27$, $P^S = 13$.

$$\pi_1^S = (13 - 4) \cdot 18 = 162, \quad \pi_2^S = (13 - 4) \cdot 9 = 81$$

Cournot comparison (same symmetric market): $BR_i = 18 - q_j/2$, $q_i^C = 12$, $P^C = 16$, $\pi_i^C = 144$.

	q_1	q_2	π_1	π_2
Cournot	12	12	144	144
Stackelberg	18	9	162	81

Firm 1 prefers Stackelberg (+18); firm 2 prefers Cournot (+63). Credible commitment also shifts surplus toward consumers: $P^S = 13 < 16 = P^C$.

General summary (1/2)

Game theory

- Dominant strategy: optimal whatever the opponent does
- Nash equilibrium: nobody deviates unilaterally
- Prisoner's dilemma: inefficient NE (collusion unstable in a one-shot game)

Bertrand (prices, simultaneous, homogeneous goods)

- $P^* = MC$, $\pi = 0$ with just 2 firms (paradox)
- Ways out: differentiation (Hotelling), capacity (Edgeworth), repeated game

Collusion and repeated games

- Sustainable in a repeated game if $\delta \geq \delta^*$ with grim trigger
- General formula: $\delta^* = (\pi^{\text{dev}} - \pi^{\text{coll}})/(\pi^{\text{dev}} - \pi^{\text{punish}})$
- δ^* is increasing in n : more firms, harder to sustain the cartel

General summary (2/2)

Cournot (quantities, simultaneous)

- BR: $q_i^*(q_j) = (a - c)/(2b) - q_j/2$
- Symmetric equilibrium: $q^C = (a - c)/[(n + 1)b]$
- Outcome between monopoly and PC; as $n \rightarrow \infty$ it converges to PC

Stackelberg (quantities, sequential)

- The leader anticipates the follower's BR: $q_1^S = (a - c)/(2b)$, $q_2^S = (a - c)/(4b)$
- First-mover advantage: $\pi_1^S > \pi_1^C > \pi_2^S$
- Requires *observable and credible commitment*

Policy lesson

Oligopoly has no single outcome: it depends on the nature of the interaction (quantities vs prices, simultaneous vs sequential, one-shot vs repeated). Antitrust authorities assess case by case, focusing on explicit collusion (cartels, such as Amica Chips/Pata/Preziosi) and abuse of dominant position. Leniency programs are a tool of applied game theory: they overturn the cartel's strategic structure in favor of competition.

Skills acquired

- ✓ Define a game in normal form, dominant strategy, and Nash equilibrium
- ✓ Find the NE in a payoff matrix using best-response highlighting
- ✓ Recognize the structure of the prisoner's dilemma and apply it to oligopoly
- ✓ Derive the Bertrand paradox and discuss its three ways out
- ✓ Compute the Cournot equilibrium (including with asymmetric costs) using reaction functions
- ✓ Compute the Stackelberg equilibrium by backward induction and compare it with Cournot
- ✓ Use geometric series to compute discounted profits over an infinite horizon
- ✓ Derive the sustainability condition for collusion (grim trigger) and compute δ^*

Optional homework exercises

1. Payoff matrix: $(C, C) = (6, 6)$, $(C, D) = (0, 10)$, $(D, C) = (10, 0)$, $(D, D) = (2, 2)$. Find the NE and comment on its Pareto efficiency. Is it a prisoner's dilemma?
2. Cournot duopoly with $P = 60 - 2Q$, $MC_1 = MC_2 = 12$. Find q_i^C , Q^C , P^C , π_i^C . Compare with monopoly and PC.
3. Same market as exercise 2. Find the Stackelberg equilibrium (firm 1 leader). Does firm 1 prefer Stackelberg or Cournot? And firm 2?
4. *3-firm cartel (the chip case)*. Same market $P = 30 - Q$, $MC = 6$, but with $n = 3$ symmetric firms (Amica, Pata, Preziosi). In case of deviation, grim trigger is activated (3-firm Cournot forever). Compute: (a) collusive profit per firm, (b) Cournot profit per firm, (c) the profit from a single deviation, (d) the minimum value of δ that sustains collusion.
5. Explain intuitively why Bertrand with two firms leads to $P = MC$ while Cournot with two firms does not. What is the key assumption that differentiates the two models?
6. *From the interest rate to sustainability*. In a certain market the cartel sustainability condition is $\delta \geq \delta^* = 0.6$. Recalling that $\delta = 1/(1+r)$ with r the annual cost of capital: (a) what is the maximum value of r compatible with collusion? (b) If the firms face a liquidity crisis and the cost of capital rises to $r = 50\%$ per year, does the cartel still hold?